

Name (Last, First):

NEU ID:

Assignment 5

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 90 = 3 + 87 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:

3 points: Correct answer with work shown

2 points: Incorrect answer but attempt shows some understanding (work shown)

1 point: Incorrect answer but an attempt was made (work shown)

0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]

(link to your live google doc)

I. The Multiplication Rule (8 x 3 = 24 pts)

1) A simple random sample of adults in a metropolitan area was selected and a survey was administered to determine political views. The results are recorded below:

Political Views				
Age	Conservative	Moderate	Liberal	Total
18-29	10	15	30	55
30-44	20	30	35	85
45-59	35	15	20	70
Over 60	20	15	10	45
Total	85	75	95	255

a) What is the probability that a person chosen at random from this sample will have moderate political views?

Answer:

b) What is the probability that a person chosen at random from those in the sample who are between the ages of 30 to 44 will have moderate political views?

Answer:

c) Based on your answers to (a) and (b), are political views and age independent for the population of adults in this metropolitan area? Why or why not?

Answer:

2) Given that $P(A) = 0.6$, $P(B) = 0.3$, and $P(B|A) = 0.5$

a) Find $P(A \text{ and } B)$

Answer:

b) Find $P(A \text{ or } B)$

Answer:

c) Are events A and B independent?

Answer:

3) At Portillo's in Rockford, the distribution of ages and what shift they work on a Saturday is below. Fill in the table with the missing information.

Age	Open - 1pm	1pm - 7pm	7pm - Close	Total
Under 18	6		21	29
18 to 49	4	5		11
40 and over	10	5		40
Total		12		80

Are the events "Open - 1pm" and "40 and over" independent? Show your work.

Answer:

II. Discrete and Continuous Random Variables (39 pts)

1) A random sample of adults was taken and asked the question “Last week, how many days did you sit down and eat at a restaurant?” X represents the number of days they responded with.

Here is the resulting probability model:

Days (X)	0	1	2	3	4	5	6	7
Probability	0.40	0.28	0.16	0.05	0.04	0.03	0.02	0.02

(a) Show that the probabilities in this distribution add up to 1.

Answer:

(b) Make a histogram of the probability distribution. Describe the shape of the distribution.

(c) Find and interpret $P(X < 4)$.

Answer:

(d) Express the event “Ate out at least twice last week” in terms of X . What is the probability of this event?

Answer:

(e) Find and interpret the mean.

Answer:

(f) Find the standard deviation.

Answer:

2) A Denver Ski Resort compiles their profit data yearly and has discovered that if it snows less than 60 inches in a season, they will lose \$90,000. If it snows between 60 inches and 80 inches, they will make \$70,000, and if it snows above 80 inches, they will make \$200,000. Let X represent the profit (any losses would be represented as a negative value). If the probability of less than 60 inches is 40%, the probability of between 60 inches and 80 inches is 45%, create a table to represent the probability distribution. Then comment on if this is a profitable business model or not based on the mean of the distribution. [6 pts]

Answer:

3) A new type of gambling game uses two dice and the sum of the two dice determines your winnings. It costs \$5 to play the game. If the sum is a 2 or a 12, the player will win \$50. If the sum is a 7, the player will win \$10. Any other sum and the player will lose and not get anything. If X represents the winnings (with -5 representing a loss, since a player will lose that amount of money), create a table to represent the probability distribution. Using the mean of this distribution, do you believe the game is in favor of the player or the "house"? [6 pts]

Answer:

4) Choose an American household at random and let the random variable X be the number of persons living in the household. The probability of X is:

X	1	2	3	4	5	6	7 or more
P(X)	0.25	0.32	0.17	0.15	0.07	0.03	0.01

Find and interpret the following probabilities:

a) $P(X > 3)$

Answer:

b) $P(X \leq 4)$

Answer:

c) $P(2 \leq x \leq 5)$

Answer:

III. Combining Random Variables (8 x 3 = 24 pts)

1) In the 2021 baseball season, the Brewers runs per game, B, varied from game to game but had $\mu_B = 3.7$ and $\sigma_B = 1.1$. The Cubs runs per game, C, were $\mu_C = 4.9$ and $\sigma_C = 1.6$.

Find the following (show your work):

μ_{B+C}	μ_{B-C}
σ_{B+C}	σ_{B-C}

2) At our town's ice cream shop, if you order a cone, you can determine the number of scoops you would like. Let X represent the number of scoops ordered and the probability that a random customer would ask for that number of scoops.

X	1	2	3	4	5
$P(X)$	0.20	0.38	0.28	0.11	0.03

It costs \$2.50 for the cone and \$0.75 for each scoop of ice cream. Find the amount of money the store expects to make on a randomly selected customer. **[6 pts]**

Answer:

3) Suppose the owners of the ice cream shop want to keep track of their profits. The cost of a cone is \$0.10 for the shop and the cost of a scoop of ice cream is \$0.45. What is the expected profit on a randomly selected customer? Use the information from problem #2 to help you answer this question. Show your work. **[6pts]**

Answer:

THE END